

Primitive Type Casting

Primitive Type Casting is the process of converting a value from one primitive data type to another primitive data type. It is used when we want to store a value of one data type into another data type.

Why Do We Need Type Casting?

Sometimes we need to:

- Store a smaller data type value into a larger data type variable.
- Store a larger data type value into a smaller data type variable.

In such cases, type casting is required.

Types of Primitive Type Casting

There are two types of type casting in Java:

1. Widening Type Casting (Implicit Casting)
2. Narrowing Type Casting (Explicit Casting)

1. Widening Type Casting

Converting a smaller data type into a larger data type automatically is called Widening Type Casting.

No extra code is required because Java performs the conversion automatically.

Characteristics

- Done automatically by Java.
- No data loss occurs.
- Small data type is converted into a larger data type.

Example

```
public class Demo {  
    public static void main(String[] args) {  
        int num = 100;  
        double value = num;  
        System.out.println(value);  
    }  
}
```

Output

100.0

Key points:

- num is an int value.
- It is stored in a double variable.
- Java automatically converts int to double.

2. Narrowing Type Casting

Converting a larger data type into a smaller data type manually is called Narrowing Type Casting.

The programmer must perform the conversion explicitly.

Characteristics

- Done manually by the programmer.
- May cause data loss.
- Larger data type is converted into a smaller data type.

Syntax

```
smallerDataType variable = (smallerDataType) value;
```

Example

```
public class Demo {  
    public static void main(String[] args) {  
        double price = 99.99;  
        int value = (int) price;  
        System.out.println(value);  
    }  
}
```

Output

99

Key points :

- price contains 99.99.
- During conversion, the decimal part is removed.
- The result becomes 99.

Comparison Between Widening and Narrowing

Widening Casting	Narrowing Casting
Automatic	Manual
Safe Conversion	Data Loss Possible
Small → Large	Large → Small
No Cast Operator Required	Cast Operator Required
int → double	double → int